

高级自然语言处理 第三次作业

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Q1:

(1)

阶段	目标	获得能力	挑战	代表模型
Pretraining	学习语言和世界知识	掌握自然语言结构和基本知识，模拟一些简单的任务	数据噪声，缺少大量高质量语料，幻觉问题，算力较昂贵	GPT-3, Gopher, BLOOM
Instruction Tuning	理解并遵循人类指令	展现推理过程，遵循指令，推广到复杂情境	需要大规模且多样的数据，需要提高代码和推理能力	InstructGPT, FLAN-T5,
Alignment Tuning	对齐人类价值与偏好	在遵守人类基本价值观前提下完成任务，拒绝有伤风化或难以完成的指令，公平正式的表达方式	成本较高，有过度对齐风险	ChatGPT, Claude, Sparrow

(2)

Emergent Abilities指的是：当模型规模（参数量或训练数据量）增加到一定临界点时，模型突然表现出一些在小模型上完全不存在、无法通过线性外推预测的新能力。例如数学推理，逻辑链式思考，上下文学习能力等。这是一种量变引起质变的表现，也是模型表示能力与任务复杂度之间的交互效应。

(3)

我在作业文件给出的github库里下载了需要的gsm8k和math的数据集，并且在示例代码的基础上进行了修改，然后分别使用Qwen2.5-0.5B进行测试，原始代码见压缩包内文件，结果如下，从上到下分别是gsm8k使用k=0，k=5；math使用k=0，k=5的结果：

```
PS C:\Users\lenovo\Desktop\advnlp\hw3> py eval_gsm8k.py --model Qwen/Qwen2.5-0.5B-Instruct --data_file gsm8k.jsonl --k_shot 0
Using k_shot = 0
The following generation flags are not valid and may be ignored: ['temperature', 'top_p', 'top_k']. Set `TRANSFORMERS_VERBOSITY=info` for more details.
Accuracy: 0.09
```

```
PS C:\Users\lenovo\Desktop\advnlp\hw3> py eval_gsm8k.py --model Qwen/Qwen2.5-0.5B-Instruct --data_file gsm8k.jsonl --k_shot 5
Using k_shot = 5
The following generation flags are not valid and may be ignored: ['temperature', 'top_p', 'top_k']. Set `TRANSFORMERS_VERBOSITY=info` for more details.
Accuracy: 0.06
```

```
Running batches: 100% | 70/70
Accuracy = 0.0
Invalid: 0
```

```
Running batches: 100% | 70/70
Accuracy = 0.014285714285714285
Invalid: 0
```

最终结果以表格形式呈现如下：

正确率	GSM8K	MATH
zero-shot CoT	0.09	0
few-shot CoT	0.06	0.014

注意到当k=5时，模型在GSM8K数据集上的正确率反而下降了，这可能是因为模型规模很小，few-shot由于上下文变长、示例分布与测试题差异等因素，反而略微降低了表现导致的。

Q2:

(1)

Diversity Filtering: LLMs 会生成大量内容，但其中许多是高度相似、冗余或模板化的。在此基础上，Diversity Filtering 的目标是减少重复，增加语义覆盖范围，提升数据集的多样性与代表性。例如ROUGE-L可以用于将 newly generated instructions 与 seed instruction 做比对，ROUGE-L 越高表示越重复，Self-Instruct 设置 ROUGE-L 小于一定的阈值才保留。

Quality Filtering: 在LLM 生成的质量参差不齐的数据中，通过评分机制筛选“自然流畅、有信息量”的样本。例如Best-of-K对同一个指令生成 K 个候选回答，随后用 reward 模型选择最佳回答。

Correctness Filtering: 一些特定的专业性强的任务（如数学、逻辑、代码等）生成的内容可能“看起来像回答”，却是错的。因此Correctness Filtering 保证只保留 事实上也正确或者逻辑上合法的样本。例如STaR的基础逻辑是如果多个 reasoning paths 中只有部分有正确答案，那么只保留正确的 reasoning chain。

(2) 我们实现了一个简单的 Reward-based Filtering pipeline：使用 Qwen2.5-0.5B生成 k=5个候选回复，随后定义了一个简单的启发式奖励函数，其中：

- length_score: 鼓励内容适中、不太短的回答。
- keyword_score: 鼓励回答覆盖预期关键术语。
- completeness_score: 避免回答中出现话没说完, 句子不完整的情况。
- repetition_penalty: 鼓励重复性较低的回答。
- fluency_score: 鼓励语言更流畅自然的回答。

并且定义reward为这五部分的加权平均:

```
return (
    0.10 * length_score +
    0.25 * keyword_score +
    0.15 * repetition_score +
    0.35 * completeness +
    0.15 * fluency
)
```

对每条指令, 保留 reward 最高的候选作为过滤后样本, 部分运行结果如图所示:

```
Candidate 2:
The attention mechanism is a key component in the Transformer architecture, which is used for tasks such as natural language processing (NLP). It works by projecting the input features into high-dimensional vectors that capture their importance. These vectors are then passed through an attention layer, which takes the projections and computes a weighted sum of them, indicating the relative importance of each feature. This allows the model to focus on important parts of the input while ignoring less relevant ones, leading to improved performance in many NLP tasks. The attention mechanism helps to overcome the vanishing gradient problem in deep neural networks, making it possible to train large models with limited computational resources.
Score = 0.8176

Candidate 3:
The attention mechanism is an important component of the Transformer architecture, which has been widely used for tasks such as natural language processing and speech recognition. Attention is a technique that allows the model to focus on different parts of the input data based on their relevance to the output, rather than relying solely on the entire input sequence. In the context of the Transformer architecture, attention is represented by a special type of matrix called an attention matrix. This matrix is initialized at the beginning of each epoch and then updated during training. At any given time, the model takes a set of inputs and computes the attention weights, which determine how much weight to give to each
Score = 0.6803

Candidate 4:
The concept of attention is central to the Transformer model, which was introduced by Hugging Face as part of the transformers library. Attention mechanisms enable the model to focus on relevant information during sequence prediction tasks such as language modeling and question answering. In a transformer model, each token or word in an input sequence has access to its own context, while also being able to see the entire input sequence. This allows the model to understand the relationships between different parts of the input and generate accurate predictions about the next element in the sequence. Attention mechanisms work by projecting the input into a higher-dimensional space that captures the most important features for each token, then weighting
Score = 0.6615

Candidate 5:
The Attention mechanism is a key component of the Transformer model, which was introduced by Vaswani et al. (2017). In the Transformer architecture, the attention mechanism allows for the efficient computation of joint and partial features between multiple inputs at different levels of abstraction. Specifically, the attention mechanism computes an attention weight matrix that represents the relative importance of each input to the output of a particular layer. This weight matrix is computed based on the dot product of the embeddings of the inputs and the weights associated with them, as well as the position of the inputs within the sequence. The attention mechanism works by projecting the inputs onto a shared embedding
Score = 0.6726

Selected: Candidate 2 (Score = 0.8176)
```

[4] Prompt: What is a language model in natural language processing?

Candidate 1:

A language model is an artificial intelligence algorithm that uses statistical models to generate human-like text. It is used for tasks such as sentiment analysis, question answering, and translation.

A language model can be trained on large amounts of text data to recognize patterns and understand the meaning behind the words. This allows it to accurately generate human-like responses to questions or prompts. Language models are often used by computer systems to process and analyze large volumes of text data, making them useful tools for various applications including speech recognition, machine translation, and more. Some examples of language models include neural networks, deep learning algorithms, and other types of artificial intelligence models.

Score = 0.7665

Candidate 2:

A language model, also known as a text model or conversational model, is a computer program that simulates human-like speech and language. It processes natural language input from humans, generates responses based on the patterns of those inputs, and can be used for tasks such as generating text, answering questions, and making recommendations.

The main components of a language model include:

1. Input: The data to be processed by the model, including text, audio, video, images, etc.
2. Hidden states: The internal state of the model, which stores information about how it has been trained and what it needs to know to generate the next

Score = 0.6832

完整代码见压缩包内文件。